Probability Mass Function and the CDF of a Discrete Variable

Contents

1. Overview . . . 2
2. From outcomes to a PMF: three coin flips . . . 3
3. The probability mass function . . . 4
   3.1 The two conditions . . . 4
4. The PMF of one die . . . 6
5. Estimating a PMF by simulation . . . 7
6. The PMF of the sum of two dice . . . 9
7. Two famous discrete distributions . . . 10
   7.1 Bernoulli distribution . . . 10
   7.2 Binomial distribution . . . 10
8. The cumulative distribution function of a discrete variable . . . 11
   8.1 Reading the CDF of two dice . . . 12
9. Summary . . . 15
10. Sources . . . 15
11. Key terms . . . 16

Where this fits: Pipeline map steps 0 (Foundations), 3 (Understand data). See the Course map.

0 Foundations	1 Frame	2 Get data	**3 Understand**	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Probability distributions; Probability density function (PDF).
- **Leads to:** Poisson distribution; Standard normal and the z-table; P-values.
- **Compare with:** Frequency tables; Likelihood.

1. Overview

Key point: The PMF gives the probability of each exact value of a discrete random variable (G-617); the CDF adds those probabilities up, giving the probability of a value at most x.

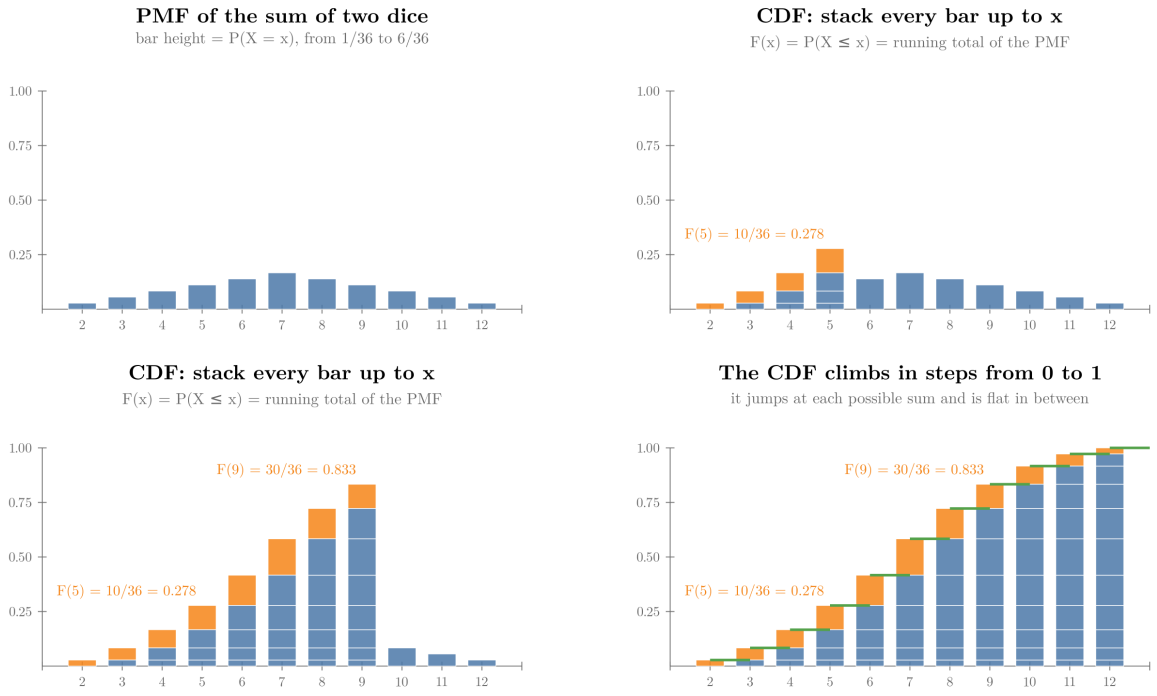


Figure 1: Building the CDF of the sum of two dice by stacking the PMF bars

Figure 1 shows the whole Note in one animation. The blue bars are the PMF of the sum of two dice: the probability of each exact sum. Stacking every bar up to x gives the CDF: the probability of a sum of x or less.

Random variables (G-1620), distributions and the PMF/PDF/CDF family are introduced in **random variables** (G-1620) and PMF, PDF and CDF. Here we build the two functions for discrete variables, in this order:

- PMF from a list of outcomes, on three coin flips (section 2);
- PMF from a formula (sections 3, 4 and 6);
- PMF from a simulation (section 5);
- two famous PMFs, Bernoulli and binomial (section 7);
- CDF: a running total of the PMF (section 8).

2. From outcomes to a PMF: three coin flips

> **Key point:** List every outcome, group the outcomes by the value of the **random variable** (G-1620), and give each value the share of outcomes in its group; the bar chart of those shares is the PMF.

Flip a fair coin three times and let X be the number of heads. X can only be 0, 1, 2 or 3: it cannot be 1.5. How likely is each value? We find out in three steps, shown in Figure 2.



Figure 2: The 8 outcomes of three coin flips are grouped by the number of heads, and each group becomes a bar of height count / 8 (idea after Khan Academy, "Constructing a probability distribution for random variable")

1. **List every outcome.** Each flip gives H or T, so three flips give 8 outcomes:

$$2 \times 2 \times 2 = 8$$

   The outcomes are HHH, HHT, HTH, HTT, THH, THT, TTH, TTT. For a fair coin all 8 are equally likely, each with probability 1/8.
2. **Write X under each outcome and group them.** One outcome has no heads (TTT), three have one head (HTT, THT, TTH), three have two heads, and one has three heads (HHH).

3. **Give each value its share.**

$$P(X = 0) = 1/8$$
$$P(X = 1) = 3/8$$
$$P(X = 2) = 3/8$$
$$P(X = 3) = 1/8$$

In the last frame of Figure 2, each group has collapsed into a bar whose height is its share. The bar chart lists every possible value of X and its probability, and there is no bar at 1.5, because $P(X = 1.5) = 0$. A list of probabilities like this, one for each value of a **discrete random variable** (G-617), is called a **probability mass function** (G-1572). The four bars add up to 1, because one of the four values is certain to happen:

$$1/8 + 3/8 + 3/8 + 1/8 = 1$$

Section 3 states the definition in general, and the rest of the Note builds PMFs for dice.

3. The probability mass function

> **Key point:** A PMF assigns a probability to each possible value of a **discrete random variable** (G-617); every probability is at least 0, and together they add up to 1.

The **probability mass function (PMF)** (G-1572) is the probability distribution function of a **discrete random variable** (G-1620). The PMF assigns a probability to each possible value:

$$p(x) = P(X = x)$$

The formula reads: "the probability that the random variable X takes the value x". For the three coin flips of section 2, the value 2 has probability:

$$p(2) = P(X = 2) = 3/8$$

On a graph, the possible values go on the x axis and their probabilities on the y axis.

3.1 The two conditions

> **Key point:** No probability is negative, and the probabilities of all possible values sum to exactly 1.

Not every function can be a PMF. Its probabilities must satisfy two conditions:

1. **In words:** no probability is negative, and the probabilities of all possible values add up to 1, because one of them is certain to happen.

2. **Formula:**

$$p(x) \geq 0 \text{ for every } x$$

$$\sum_x p(x) = 1$$

   Here $\sum_x$ means: add $p(x)$ over every possible value x.
3. **Example:** for one die, every $p(x)$ is $1/6$, which is at least 0, and the six values add up to 1:

$$\sum_{x=1}^{6} p(x) = 6 \times \frac{1}{6}$$

$$\sum_{x=1}^{6} p(x) = 1$$

   For the sum of two dice (the table in the sum of two dice):

$$1 + 2 + 3 + 4 + 5 + 6 = 21$$

$$5 + 4 + 3 + 2 + 1 = 15$$

$$\frac{21 + 15}{36} = \frac{36}{36} = 1$$

Figure 3 stacks these probabilities into one column for each experiment. Watch the top: six slices of $1/6$, or eleven slices from $1/36$ to $6/36$, both fill exactly the height 1.

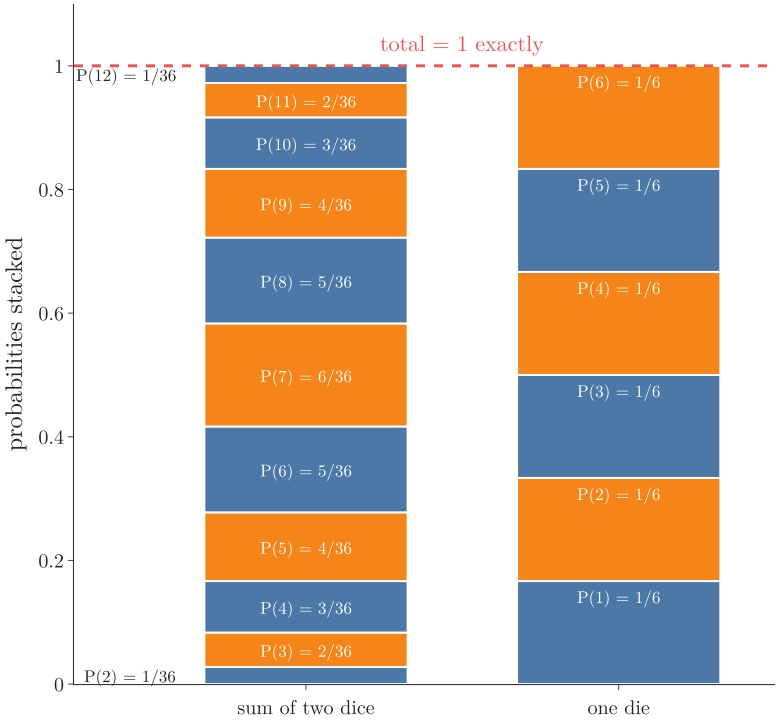


Figure 3: The probabilities of one die and of the sum of two dice, stacked: each PMF fills exactly 1

4. The PMF of one die

> **Key point:** For a fair die the PMF is 1/6 at each face from 1 to 6 and 0 everywhere else.

A PMF written as a formula $y = f(x)$ must give a value for **every** x, including values the die cannot show. So it has two parts. One new sign appears: $\in$ reads "is one of" or "belongs to". For example, $3 \in \{1, 2, 3, 4, 5, 6\}$ is true (3 is a face of the die), while $7 \in \{1, 2, 3, 4, 5, 6\}$ is false. The curly brackets { } list the members of a set.

$$f(x) = \begin{cases} \dfrac{1}{6} & \text{if } x \in S \\ 0 & \text{otherwise} \end{cases}$$

Here S is the set of faces:

$$S = \{1, 2, 3, 4, 5, 6\}$$

The formula says at once that every face is equally likely, and that 1.5 or 7 have probability 0. Its graph is six equal bars; a distribution in which every value is equally likely is called a **discrete uniform distribution** (G-618). Figure 4 draws the formula along the whole x axis: six spikes of height 1/6, and 0 everywhere else, including at 1.5 and 7.

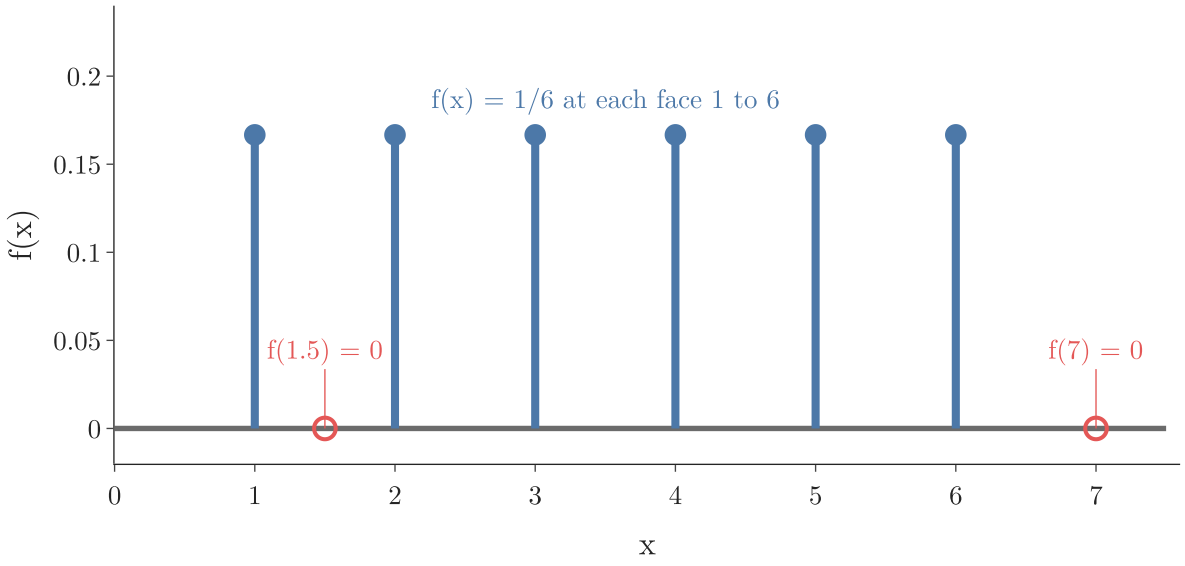


Figure 4: The PMF of one die as a function of every x: 1/6 at the six faces, 0 elsewhere

5. Estimating a PMF by simulation

> **Key point:** Repeat the experiment many times, count each outcome and divide by the number of trials; the shares approach the true PMF.

We can also find a PMF without any formula: run the experiment many times and count. The share of trials that gave each value is an estimate of its probability, the **estimated PMF** (G-708), the same idea as the **relative frequency** (G-1664) of a frequency table.

1. **In words:** the estimated probability of a value is the number of times it appeared divided by the number of trials.
2. **Formula:**
$$\hat{p}(x) = \frac{\text{count of } x}{\text{number of trials}}$$
3. **Example:** in 10,000 simulated rolls of one die, the face 1 appeared 1,681 times:
$$\hat{p}(1) = \frac{1681}{10000} = 0.1681$$
The exact value is $1/6 \approx 0.1667$.

The hat on $\hat{p}$ marks an estimate. Figure 5 compares the simulated PMFs (bars) with the exact ones (diamonds), for one die and for the sum of two dice.

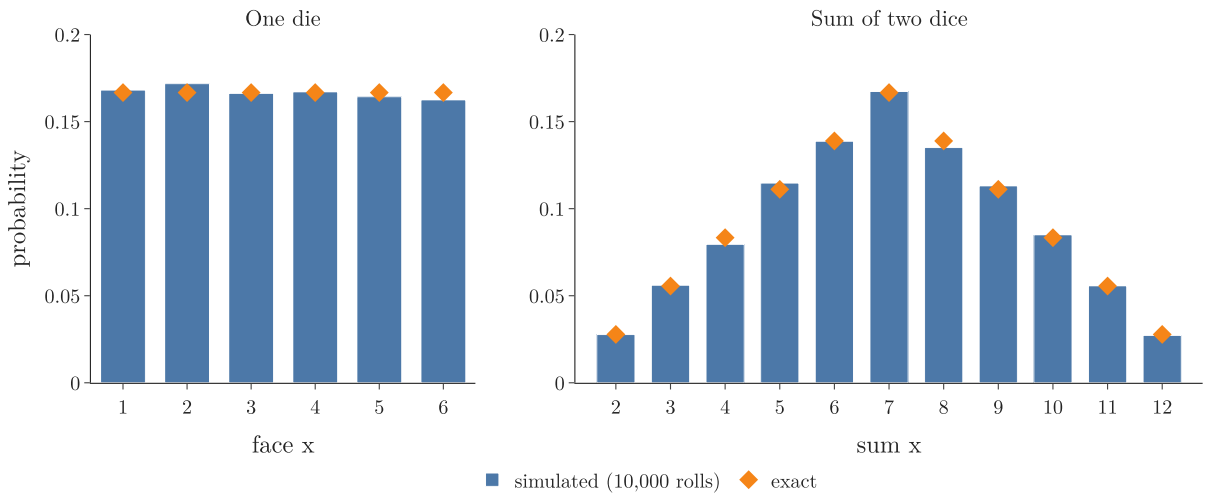


Figure 5: PMFs estimated from 10,000 simulated rolls (bars) and the exact PMFs (diamonds)

The estimates are close but not exact: the six faces came out between 0.1625 and 0.1718, around 1/6. A random experiment rarely matches theory exactly. For the sum of two dice, the simulated bars follow the same rise to 7 and fall after it as the exact values.

> **Python:** Simulating a die and estimating its PMF.

```
import numpy as np
```

```
import pandas as pd

rng = np.random.default_rng(42)        # fixed seed
rolls = rng.integers(1, 7, size=10_000)
# count each face, divide by the total
pmf = pd.Series(rolls).value_counts(normalize=True)
pmf = pmf.sort_index()
pmf.plot(kind="bar")
```

`integers(1, 7)` excludes the upper end, so it returns 1 to 6. `normalize=True` divides the counts by their total; without it we would divide by 10,000 ourselves. `sort_index()` puts face 1 first. For two dice, add two such arrays. The Notebook (`MA-021-pmf-and-discrete-cdf.ipynb`) runs both experiments.

Extra: The more trials, the closer the estimate gets to the true probability. The pull of the estimate towards the true probability is the **law of large numbers** (G-1052; Pishro-Nik §7.1.1). With 100 rolls a face typically comes out anywhere from about 0.12 to 0.22; with our 10,000 rolls all six land within about 0.005 of 1/6 (the Notebook repeats the 100-roll experiment 1,000 times).

Figure 6 watches the same 10,000 rolls arrive. Watch the title: after 10 rolls face 1 is 0.133 away from 1/6; after all 10,000, no face is more than about 0.005 away.

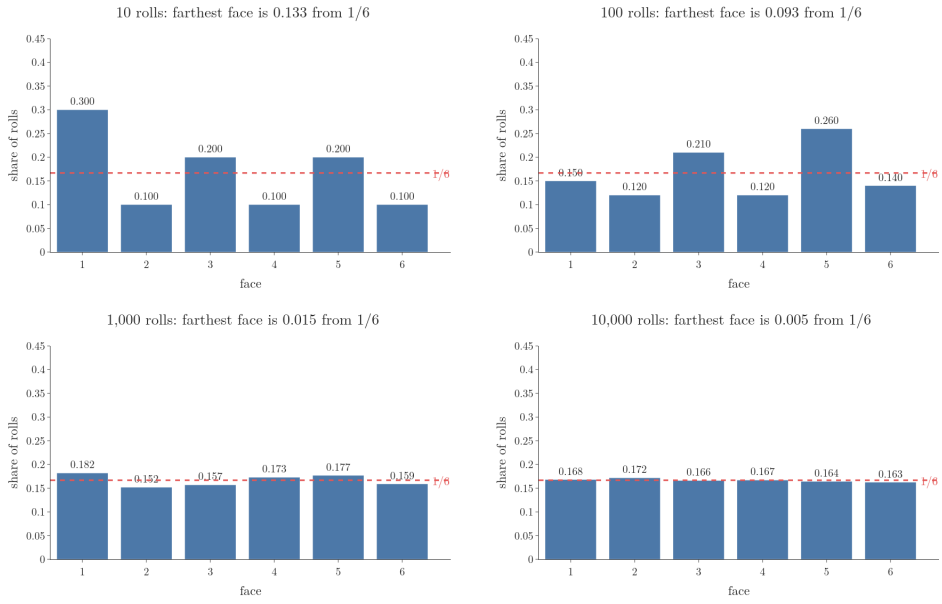


Figure 6: The estimated PMF of one die after 10, 20, 50, . . . , 10,000 of the same simulated rolls, against the exact 1/6

6. The PMF of the sum of two dice

> **Key point:** When the probabilities differ, the PMF formula must give each value its own probability; for two dice one line does it.

The probabilities of the sum of two dice, from 1/36 for sums 2 and 12 up to 6/36 for 7, are counted in the sum of two dice and plotted in Figure 5 (right). Written as a PMF, the list becomes a formula.

> **Extra:** The number of pairs falls by one for each step away from 7, so:
>
> 1. **In words:** start from 6 pairs at the sum 7 and subtract the distance from 7.
> 2. **Formula:** for $x = 2, \ldots, 12$,
>
> $$f(x) = \frac{6 - |x - 7|}{36}$$
>
> 3. **Example:** for $x = 9$, the distance from 7 is 2:
>
> $$f(9) = \frac{6 - 2}{36} = \frac{4}{36} \approx 0.111$$

Figure 7 shows the rule as a shape: the pair counts form a tent, peaking at 6 for the sum 7 and losing one pair per step away from it.

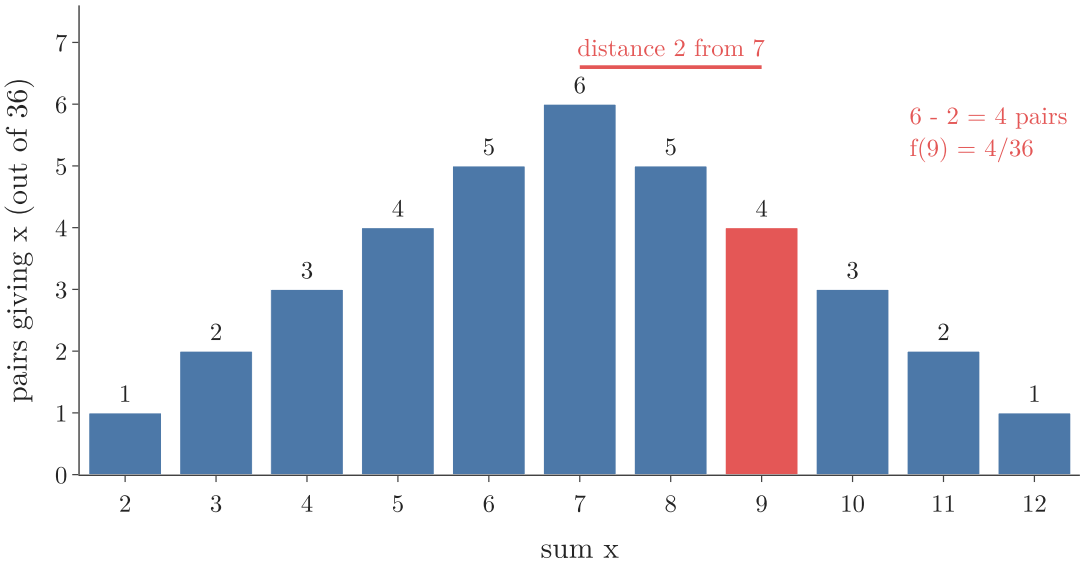


Figure 7: Pairs of dice giving each sum: 6 minus the distance from 7, such as 4 pairs for the sum 9

7. Two famous discrete distributions

> **Key point:** The Bernoulli PMF describes one yes/no trial; the binomial PMF counts the successes in n such trials.

Many discrete experiments follow a famous PMF (see famous distributions). Two of them come up constantly, Bernoulli and binomial, taught in full in the Bernoulli distribution and the binomial distribution.

7.1 Bernoulli distribution

> **Key point:** One trial with two outcomes: 1 with probability p, 0 with probability $1 - p$.

The **Bernoulli distribution** (G-275) describes a single trial with two outcomes, success ($k = 1$) and failure ($k = 0$), such as one coin toss.

1. **In words:** success has probability p; failure has the rest, $q = 1 - p$.
2. **Formula:**
$$P(X = k) = \begin{cases} p & \text{if } k = 1 \\ q = 1 - p & \text{if } k = 0 \end{cases}$$
3. **Example:** with $p = 0.3$, $P(X = 1) = 0.3$ and $P(X = 0) = 0.7$; they add up to 1.

The Bernoulli distribution has **one** parameter, p (with $0 \leq p \leq 1$). The q often listed beside it is not a second parameter: it is fixed by p as $1 - p$. (MML Example 6.8)

7.2 Binomial distribution

> **Key point:** The number of successes in n independent Bernoulli trials with the same p.

The **binomial distribution** (G-308), used in why voting works, counts the successes in n independent trials, each with success probability p. Its parameters are n and p.

1. **In words:** choose which k of the n trials succeed, then multiply the probabilities of k successes and $n - k$ failures.
2. **Formula:**
$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$
where $\binom{n}{k}$, read as n choose k, is the number of ways to choose k trials out of n.
3. **Example:** the probability of exactly 2 heads in 4 fair coin tosses ($n = 4$, $p = 0.5$). There are $\binom{4}{2} = 6$ ways to place the 2 heads:
$$P(X = 2) = 6 \times 0.5^2 \times 0.5^2$$
$$P(X = 2) = 6 \times 0.0625$$
$$P(X = 2) = 0.375$$

Figure 8 draws both PMFs: Bernoulli has just two bars, and the binomial for 4 tosses has five, with 2 heads the most likely at 0.375.

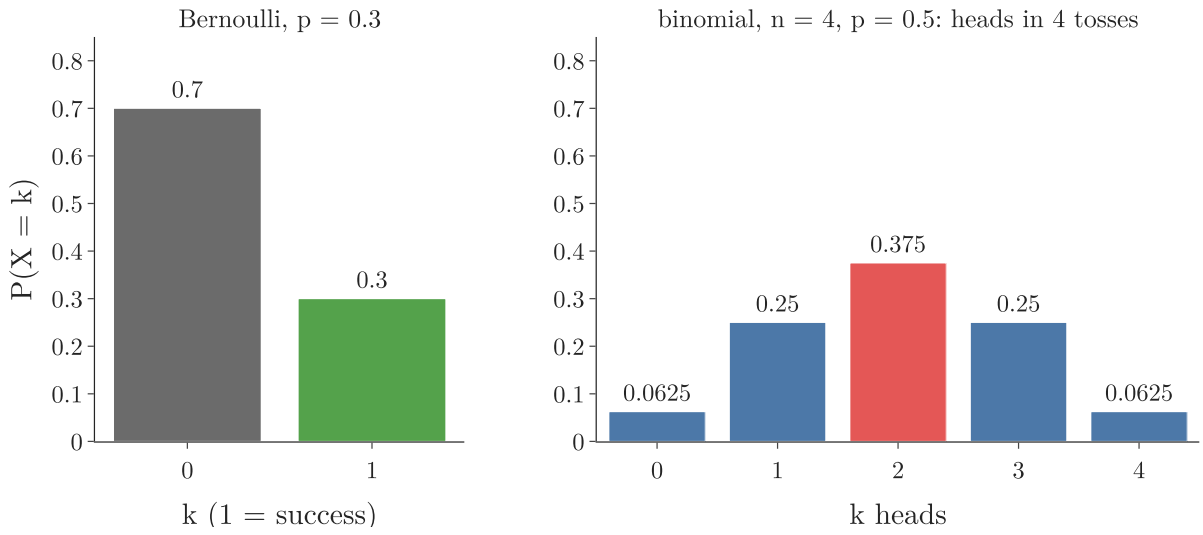


Figure 8: The Bernoulli PMF with p = 0.3 (left) and the binomial PMF of heads in 4 fair tosses (right)

8. The cumulative distribution function of a discrete variable

> **Key point:** The CDF $F(x)$ is the probability that the variable is at most x: the running total of the PMF up to x.

The **cumulative distribution function (CDF)** (G-515) of a **random variable** (G-1620) X gives the probability that X takes a value **less than or equal to** x:

$$F(x) = P(X \leq x)$$

The difference from the PMF is one symbol:

- **PMF at 4:** $P(X = 4)$, the probability of **exactly** 4.
- **CDF at 4:** $P(X \leq 4)$, the probability of 4 **or less**: 1, 2, 3 or 4.

For a discrete variable, "4 or less" means adding the PMF of every value up to 4. That running total is why the function is called cumulative.

1. **In words:** add the probabilities of all possible values up to and including x.
2. **Formula:**

$$F(x) = \sum_{t \leq x} p(t)$$

   Here $\sum_{t \leq x}$ adds $p(t)$ over every possible value t that is at most x.

3. **Example:** for one die,

$$F(4) = p(1) + p(2) + p(3) + p(4)$$

$$F(4) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$$

$$F(4) = \frac{4}{6} \approx 0.667$$

The CDF of the die climbs $1/6, 2/6, 3/6, 4/6, 5/6, 6/6$: it starts above 0 and ends at exactly 1.

The discrete CDF is the **cumulative relative frequency** (G-518) of a frequency table, with probabilities in place of relative frequencies.

8.1 Reading the CDF of two dice

> **Key point:** The CDF answers "what is the chance of x or less?" in one reading: a sum of 9 or less has probability 30/36, about 0.83.

Adding up the two-dice PMF gives its CDF:

Sum x	$P(X = x)$	$F(x) = P(X \leq x)$
2	1/36	$1/36 = 0.028$
3	2/36	$3/36 = 0.083$
4	3/36	$6/36 = 0.167$
5	4/36	$10/36 = 0.278$
6	5/36	$15/36 = 0.417$
7	6/36	$21/36 = 0.583$
8	5/36	$26/36 = 0.722$
9	4/36	$30/36 = 0.833$
10	3/36	$33/36 = 0.917$
11	2/36	$35/36 = 0.972$
12	1/36	$36/36 = 1$

Figure 9 draws both functions on the same x axis. On the PMF we read the chance of exactly x; on the CDF, the chance of x or less:

- A sum of 9 or less happens in about 83% of rolls:

$$F(9) = 30/36 \approx 0.83$$

- A sum of 5 or less happens in about 28%:

$$F(5) = 10/36 \approx 0.28$$

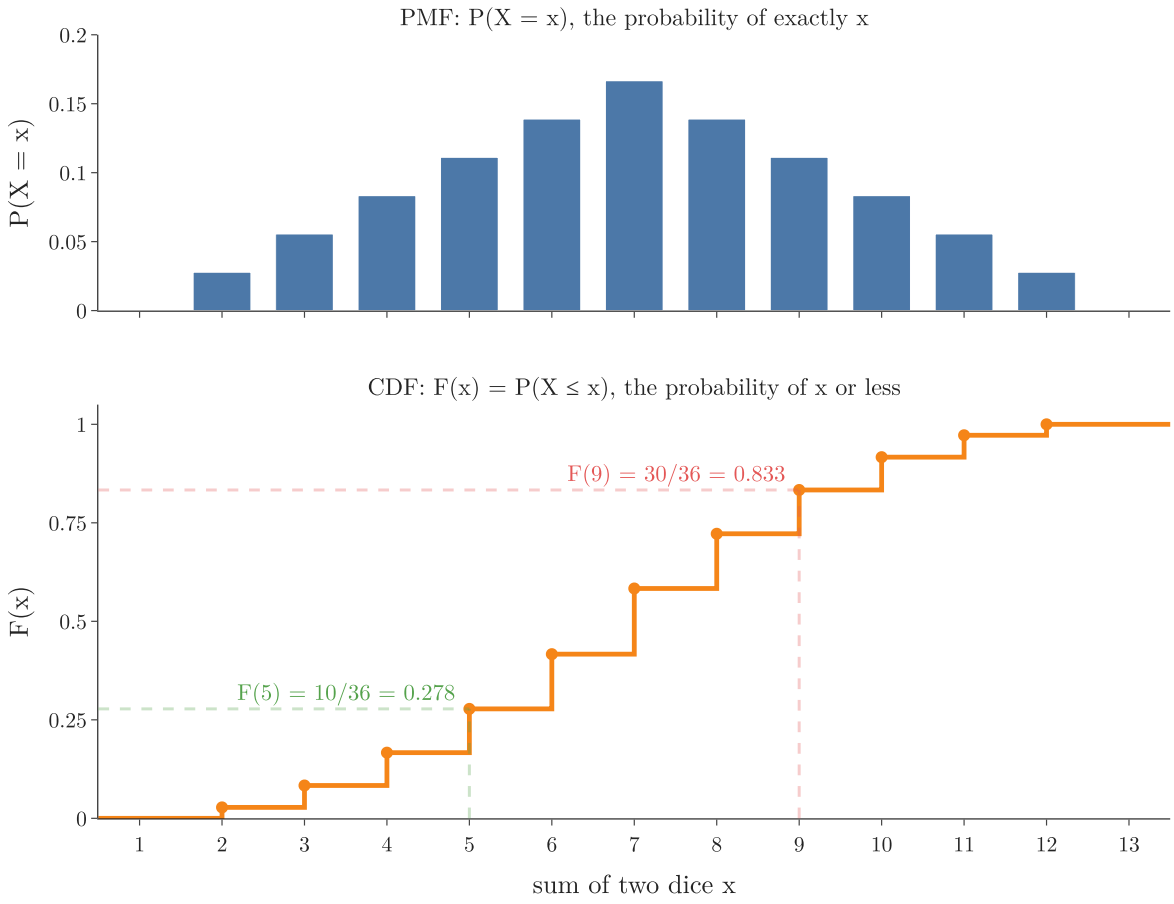


Figure 9: PMF (top) and CDF (bottom) of the sum of two dice

Python: The CDF is the cumulative sum of the PMF.

```
cdf = pmf.cumsum()          # pandas
cdf = np.cumsum(pmf)        # NumPy, same values
```

The simulated CDF from the 10,000 rolls gives $F(9) = 0.832$, against the exact 0.833.

There is no separate name such as "cumulative mass function": the CDF is called the same for discrete and continuous variables.

Extra: Three facts about every CDF, and two uses of them:

- **A CDF runs from 0 to 1 and never goes down.** Each step adds a probability, which is never negative.
- **A CDF is defined for every x, not only the possible values.** No sum lies between 7 and 7.5, so:

$$F(7.5) = P(X \leq 7.5)$$

$$F(7.5) = P(X \leq 7) = 21/36$$

So the CDF of a discrete variable is a **step function** (G-1889): flat between possible values, jumping up at each one by that value's probability (Figure 9, bottom).

Drawing it as bars, one per value, shows only its values at the jumps.

- **The PMF can be read back from the jumps:** the jump at 9 is the difference of two CDF values:

$$F(9) - F(8) = 30/36 - 26/36$$

$$F(9) - F(8) = 4/36 = p(9)$$

- **Probability of a range:** $P(a < X \leq b) = F(b) - F(a)$. For a sum above 5 and at most 9:

$$P(5 < X \leq 9) = \frac{30}{36} - \frac{10}{36}$$

$$P(5 < X \leq 9) = \frac{20}{36} \approx 0.556$$

- **Probability of more than x:** $P(X > x) = 1 - F(x)$. A sum above 9 has probability:

$$1 - F(9) = 1 - 30/36$$

$$1 - F(9) = 6/36$$

9. Summary

Function	Question it answers	One die	Two dice	Why it matters
PMF $p(x) = P(X = x)$	chance of exactly x	$p(4) = 1/6$	$p(9) = 4/36$	a discrete variable has separate values, so each can carry its own probability
CDF $F(x) = P(X \leq x)$	chance of x or less	$F(4) = 4/6$	$F(9) = 30/36$	gives the chance of x or less in one reading, and a range by subtracting two values
Estimated PMF $\hat{p}(x)$	share of trials giving x	$\hat{p}(1) = 0.1681$	$\hat{p}(7) = 0.1673$	needs no formula, only trials

- A PMF needs $p(x) \geq 0$ and $\sum p(x) = 1$, because no probability is negative and one of the values is certain to happen.
- A PMF formula covers every x: it is 0 for impossible values, so a die's PMF gives 1.5 and 7 the probability 0.
- Simulating many trials and dividing counts by the number of trials estimates the PMF; more trials bring the estimate closer to the true PMF (the law of large numbers).
- Bernoulli: one trial, one parameter p. Binomial: successes in n trials, parameters n and p. Both come up constantly; the binomial is behind why voting works.
- The CDF is the running total of the PMF: a **step function** (G-1889) from 0 to 1, because it jumps by each value's probability at that value and stays flat between values.

10. Sources

Built from

- CampusX, "Session 40 - Probability Distribution Functions - PDF, PMF & CDF | DSMP 2023", YouTube, https://www.youtube.com/watch?v=C_QAURbgBqY
- Khan Academy, "Constructing a probability distribution for random variable", YouTube, https://www.youtube.com/watch?v=cqK3uRoPtk0

Other references

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Example 6.8 (Bernoulli distribution).
- Pishro-Nik, H. (2014). *Introduction to Probability, Statistics, and Random Processes*. Kappa Research. Section 7.1.1 (law of large numbers).

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Probability mass function (PMF) (G-1572)	The function that gives the probability of each exact value of a discrete random variable, such as $P(X = 3) = 1/6$ for a fair die; its probabilities add up to 1.
Discrete uniform distribution (G-618)	A discrete distribution in which every possible value is equally likely, such as a fair die.
Estimated PMF (G-708)	Each value's share of many repeated trials, used as its probability.
Law of large numbers (G-1052)	The more trials, the closer a share of trials gets to the true probability.
Bernoulli distribution (G-275)	The distribution of a single trial with two outcomes, such as one coin toss: 1 (success) with probability p, 0 (failure) with probability $1 - p$.
Cumulative distribution function (CDF) (G-515)	For every value x, the probability that the random variable comes out at most x: $P(X \leq x)$.
$\binom{n}{k}$ (n choose k) (G-7)	The number of ways to choose k items out of n.
$P(X = x)$ (G-34)	The probability that the random variable X takes the value x.
Discrete random variable (G-617)	A random variable that takes separate values, such as a die's face.
Random variable (G-1620)	The possible numerical outcomes of a random experiment; strictly, a function from outcomes to numbers.
Relative frequency (G-1664)	A category's share of all the data: its frequency divided by the total.
Binomial distribution (G-308)	The distribution of the number of successes in n independent trials with the same success probability.
Cumulative relative frequency (G-518)	The running total of the relative frequencies; ends at 1.
Step function (G-1889)	A function that outputs 1 for positive inputs and 0 otherwise; the classic perceptron uses it to turn its score into a class.